MACHINE LEARNING ALGORITHMS / MIDTERM 2026

Can metadata predict how long a game takes?

Regression analysis of main-story completion time using structured video-game data

IGDB-BASED DATA

FIXED TEST SPLIT

ACTUAL CLEANED GAMES

5,229

each with a positive main-story time

COMPLETION TIME PERCENTILES

P25	MEDIAN	P95
2 h	5 h	40 h

Measured data, not synthetic

Direction 42 · Video games · 08 Oct 2026

What are we trying to predict?

A practical continuous target—not a review label or a game genre.

RESEARCH QUESTION

How accurately can a game's genre, developer profile and release metadata predict its main-story completion time?

STRUCTURED INPUTS

- Primary genre
- All genres
- Developer catalog tier
- Release year
- Price unavailable
- Tags unavailable

TARGET

main_story_hours

A positive numerical value, measured in hours

REGRESSION → HOURS

Descriptions are preserved in the dataset—but excluded from midterm model inputs.

DIRECTION 42 / DATA AUDIT

5,229 usable games after cleaning

Collected on 8 October 2026; one row describes one game.

5,273	→	−34	→	−10	→	5,229
RAW RECORDS		INVALID / INCOMPLETE		DUPLICATES		CLEAN RECORDS

AVAILABLE FOR MODELING

- **Genre + genre groups**
  24 primary genres
- **Developer identity**
  2,488 unique developers
- **Release year**
  ~100% populated

MISSING FIELDS

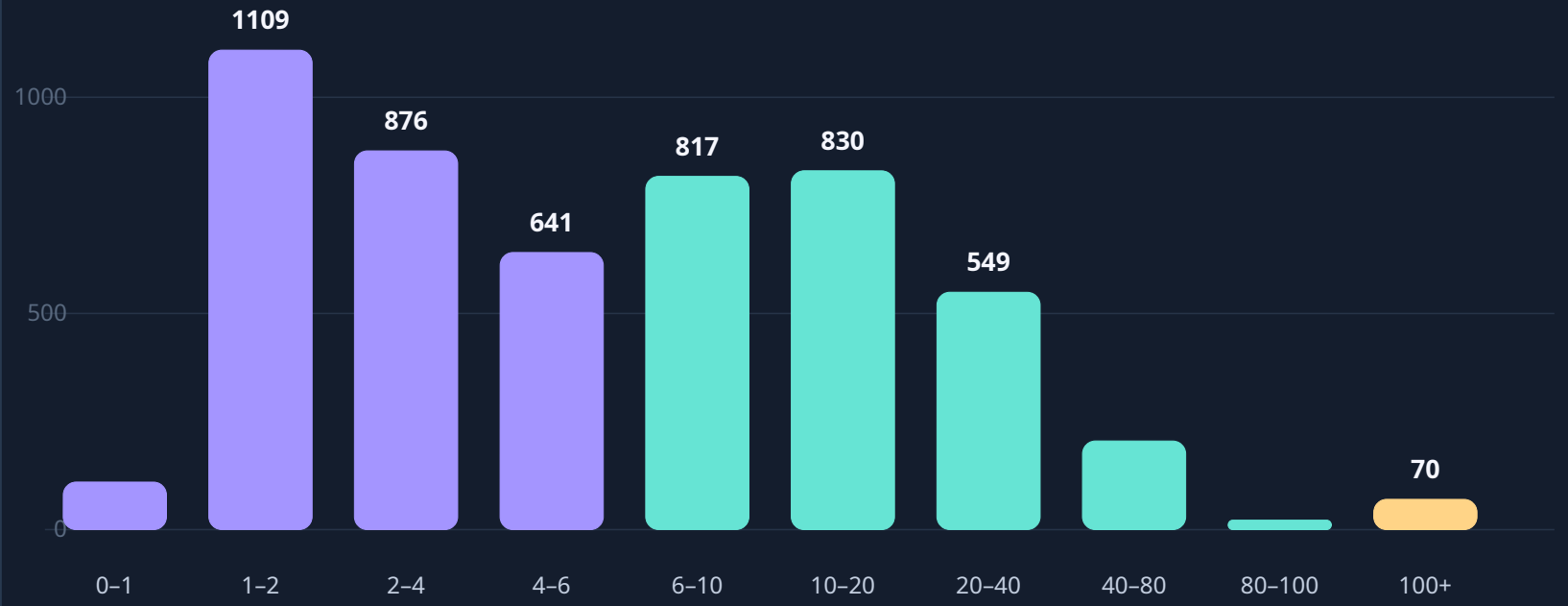
Price: 100% missing
Tags: 100% empty

Dataset provenance in notebook: IGDB API game-time-to-beat endpoint + optional Steam metadata.

Most games are short; a few take hundreds of hours

A strongly right-skewed target makes squared-error metrics especially sensitive to outliers.

GAMES BY MAIN-STORY DURATION



Duration range in hours · final bin includes 100 h and above

PERCENTILES

25th	2 h
50th	5 h
75th	12 h
95th	40 h
99th	100 h

Longest recorded: 1,000 h

A fixed split and training-only preprocessing

Model selection used three-fold cross-validation on the training partition.

01

LOCK DATA

5,229 game IDs

Raw snapshot; no target imputation

→

02

SPLIT ONCE

4,183 train / 1,046 test

80 / 20 split · random seed 42

→

03

TRANSFORM IN FOLDS

Feature encoding + imputation

Scale numbers; encode categories

→

04

CHOOSE ON CV

Lowest three-fold MAE

Only then evaluate the held-out test

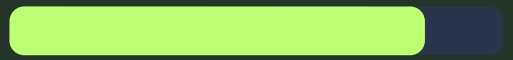
FIVE MIDTERM MODELS

Median dummy | Linear | **Ridge** | Decision tree | KNN

The target is log1p-transformed for fitting and converted back into hours for evaluation. Descriptions are excluded.

Ridge has the lowest training CV MAE

Five candidates were compared using the same held-out test partition. Lower is better.

MODEL	TRAIN CV MAE	CV HOURS	TEST MAE	TEST R^2
Median baseline		11.07	13.01	-0.041
Linear regression		10.12	11.83	-0.005
Ridge regression		**10.07**	11.82	-0.012
Decision tree		10.19	11.81	+0.006
K-nearest neighbors		10.40	11.91	-0.006

Chosen by training CV—not by the marginally lower test MAE of the decision tree.

A modest gain in MAE—not strong explanatory power

The preselected Ridge model is better than the median dummy on the 1,046 held-out games.

SELECTED RIDGE MODEL

11.82 h

HELD-OUT MEAN ABSOLUTE ERROR

Median baseline 13.01 h

↓ 9.2% lower MAE

3.20 h	51.43 h	−0.012
MEDIAN ABSOLUTE ERROR	RMSE	R^2 ON TEST

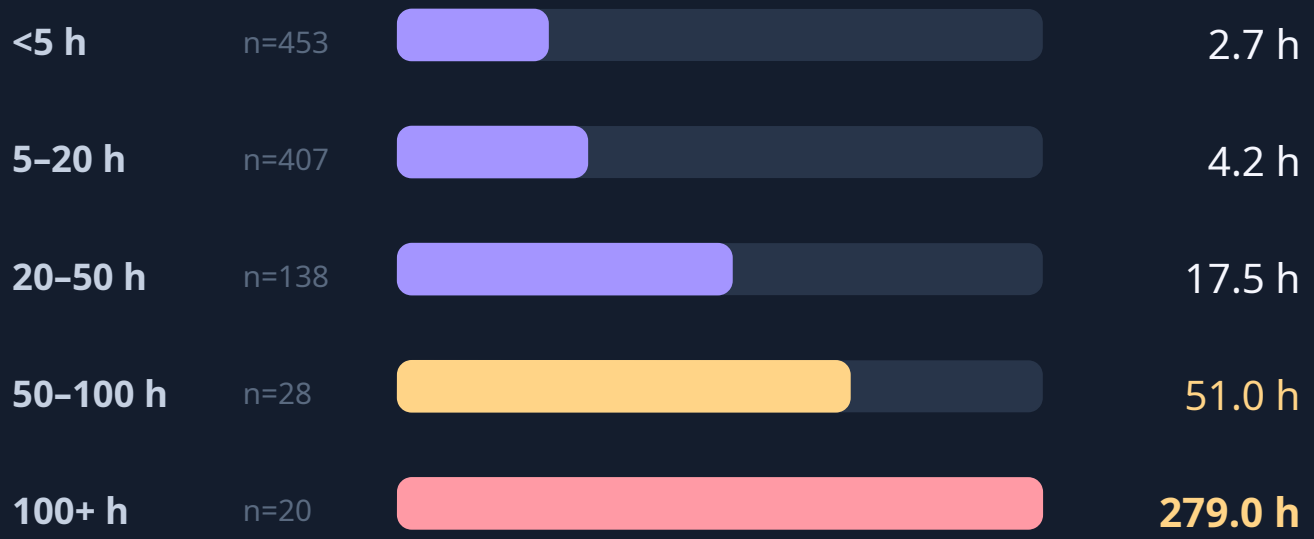
WHAT THIS MEANS

Ridge improves the primary MAE goal, but its negative R^2 and high RMSE reveal large remaining errors—especially on very long games.

Long-running games dominate the errors

Held-out absolute error rises sharply as actual completion time increases.

TEST ERROR BY ACTUAL GAME LENGTH



Bar lengths shown on logarithmic scale; values are MAE in actual hours.

LARGEST MISSES / TEST SET

Gorilla Tag

998 h actual → 1.4 h predicted

Grand Theft Auto Online

513 h actual → 7.8 h predicted

Rainbow Six Siege

500 h actual → 6.4 h predicted

Check whether time-to-beat targets are meaningful for ongoing online games.

The baseline is established; substantial work remains

Measured midterm answer: modest MAE improvement, limited predictive reliability.

WHAT WE CAN DEFEND

01 **5,229 cleaned games**

A reproducible dataset and held-out split

02 **9.2% lower test MAE**

Ridge vs the median dummy baseline

03 **R^2 remains below zero**

Outliers and missing metadata limit accuracy

WHAT COMES NEXT

ENDTERM

Ensemble tuning · game clustering · neural network

FINAL

TF-IDF descriptions, structured-vs-text ablation, paired evaluation and a local prediction app

Next quality fix: verify long-duration labels and recover useful prices/tags.